

YEAR 7 MATHEMATICS

Chapter 5A

Algebra

MIXED PRACTICE WORKSHEETS

5A

How to use this booklet

- 1 Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- 2 Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- 3 Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- 4 Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

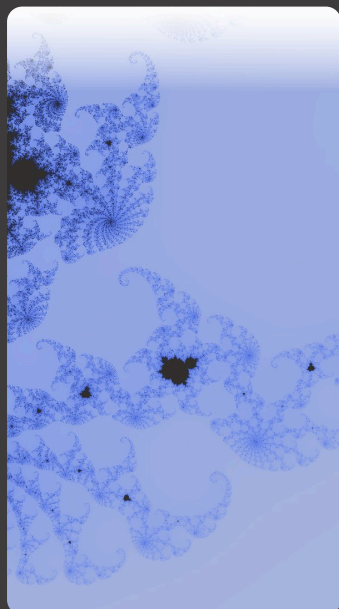
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
5.01	The commutative and associative laws	3		☐
5.02	The distributive law	5		☐
5.03	Variables and algebraic notation	7		☐
5.04	Expressions from words	9		☐
5.05	Substitution	11		☐
5.07	Adding and subtracting terms	13		☐
5.08	Multiplying terms	15		☐
5.09	Dividing terms	17		☐

In this chapter you will

- use the commutative, associative and distributive laws
- use variables and algebraic notation
- write expressions from words
- substitute into expressions and formulas
- add, subtract, multiply and divide algebraic terms



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is Elephant Valley, the row of trunks along the right-hand edge of the main body. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

5.01 The commutative and associative laws

YEAR 7 · CHAPTER 5A: ALGEBRA · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

You can add or multiply numbers in any order (the commutative law) and group them any way (the associative law). So pair up friendly numbers first: $25 \times 4 = 100$.

Watch your teacher.
Copy every step.

E1 $25 \times 17 \times 4$

E2 $38 + 47 + 62$

E3 Is $20 - 8$ the same as $8 - 20$?

Easy

Questions 1–13

1 $14 \times 20 \times 5$ 2 $67 + 52 + 3$

3 Write 11×6 another way. Is the answer the same?

4 $2 \times 9 \times 50$ 5 $86 + 22 + 4$

6 Write 8×12 another way. Is the answer the same?

7 $25 \times 3 \times 4$ 8 $11 + 58 + 9$

9 What is 10% of 651?

from 4.11

10 Write 10×3 another way. Is the answer the same?

11 $2 \times 23 \times 50$ 12 $72 + 31 + 8$

13 Write 8×5 another way. Is the answer the same?

Easy

Questions 14–22

14 $47 + 42 + 3$

15 Write 3×10 another way. Is the answer the same?

16 $50 \times 3 \times 2$

17 $700 \div 100$

from 3.03

18 $125 \times 12 \times 8$

= $___ \times ___ \times ___$

= $___ \times ___$

= $___$

19 $77 + 65 + 3$

20 Write 9×6 another way. Is the answer the same?

21 Use a calculator to write $\frac{13}{50}$ as a decimal.

from 4.09

22 $27 \times 20 \times 5$

Medium

Questions 23–30

23 $125 \times 6 \times 8$

24 $26 + 24 + 74 + 25$

25 Which is larger: $\frac{1}{3}$ or $\frac{1}{12}$?

from 4.02

26 Is $(24 \div 4) \div 2$ the same as $24 \div (4 \div 2)$?

27 $5 \times 18 \times 2$

28 $\sqrt{49} + \sqrt{81}$

from 3.08

29 $23 + 31 + 77 + 83$

30 Is 7×6 the same as 6×7 ?

5.01 The commutative and associative laws (continued)

YEAR 7 · CHAPTER 5A: ALGEBRA ·
PAGE 2 OF 2

Medium

Questions 31–40

31 $37 + 68 + 63 + 75$

32 Is $(10 - 4) - 3$ the same as $10 - (4 - 3)$?

33 Round 49 to its first digit.

from 3.01

34 $50 \times 18 \times 2$

35 Write 0.2 as a percentage and as a fraction.

from 4.10

36 $5 \times 6 \times 6 \times 2$

37 $31 + 45 + 69 + 21$

38 How many $\frac{1}{5}$ s are in 2? Work out $2 \div \frac{1}{5}$.

from 4.07

39 Is $(8 + 5) + 2$ the same as $8 + (5 + 2)$?

40 $3\frac{1}{2} \times \frac{2}{5}$

from 4.06

Challenging

Questions 41–50

41 $20 \times 6 \times 8 \times 5$

42 Which law:
 $(6 + 8) + 2 = 6 + (8 + 2)$?

43 Write $2 \times 3 \times 3 \times 3$ in index form.

from 3.10

44 Evaluate $15 \times 14 \times 2$ in your head. Show the order you used.

45 Is 9 228 divisible by 4?

from 3.04

46 $101 + 55 + 99 + 45$

47 $25 \times 3 \times 5 \times 4$

48 Write $1\frac{1}{2}$ as an improper fraction.

from 4.01

49 Which law: $4 \times 15 = 15 \times 4$?

50 Use the HCF to simplify the fraction $\frac{54}{72}$.

from 3.11

Challenging

Questions 51–60

51 Which law:
 $(5 \times 2) \times 7 = 5 \times (2 \times 7)$?

52 Evaluate $25 \times 44 \times 2$ in your head. Show the order you used.

53 Write 4 out of 6 as a fraction in simplest form.

from 4.12

54 $132 + 79 + 68 + 21$

55 List the primes between 10 and 30.

from 3.09

56 $25 \times 19 \times 2 \times 4$

57 Is $12 \div 4$ the same as $4 \div 12$?

58 $\frac{4}{7}$ of 49

from 4.05

59 Which law: $9 + 13 = 13 + 9$?

60 Write CXVI as a number.

from 3.06

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

The distributive law: multiply each part of the bracket. $6 \times 34 = 6 \times 30 + 6 \times 4$. It works with subtraction too: $9 \times 34 = 10 \times 34 - 1 \times 34$.

Watch your teacher.
Copy every step.

E1 7×43

E2 9×26

E3 Expand $5(x + 3)$.

Easy

Questions 1–9

1 4×34

$= \underline{\quad} + \underline{\quad} = \underline{\quad}$

2 Work out $9 \times (50 + 9)$ two ways.

3 Expand $4(m + 12)$.

4 8×43

5 Work out $8 \times (50 + 1)$ two ways.

6 Expand $2(x + 4)$.

7 3×34

8 Work out $3 \times (40 + 5)$ two ways.

9 $67 + 40 + 3$

from 5.01

Easy

Questions 10–18

10 Work out $6 \times (30 + 3)$ two ways.

$\underline{\quad} \times \underline{\quad} = \underline{\quad}$

$\underline{\quad} + \underline{\quad} = \underline{\quad}$

11 Expand $4(n + 3)$.

12 5×29

13 Which is larger: $\frac{1}{12}$ or $\frac{1}{7}$?

from 4.02

14 19×46

15 Work out $3 \times (10 + 8)$ two ways.

16 Expand $9(x + 10)$.

17 Write 9×5 another way. Is the answer the same?

from 5.01

18 7×47

Medium

Questions 19–26

19 19×34

20 Expand $3(c - 6)$.

21 $20 \times 4 \times 5$

from 5.01

22 Fill in the box:
 $7 \times 27 = 7 \times 20 + 7 \times \square$

23 29×30

24 $8\frac{4}{5} - 2\frac{3}{5}$

from 4.04

25 Expand $5(t - 10)$.

26 Fill in the box:
 $5 \times 25 = 5 \times 20 + 5 \times \square$

Medium

Questions 27–37

27 Expand $2(k - 7)$.

28 Fill in the box:
 $4 \times 36 = 4 \times 30 + 4 \times \square$

29 $56 + 71 + 4$

from 5.01

30 9×18

31 Is $(24 \div 4) \div 2$ the same as $24 \div (4 \div 2)$?

from 5.01

32 Expand $3(3a + 4)$.

33 Expand $2(y - 9)$.

34 Simplify $\frac{48}{52}$ fully.

from 4.01

35 Fill in the box:
 $4 \times 61 = 4 \times 60 + 4 \times \square$

36 $23 \times 50 \times 2$

from 5.01

37 9×59

Challenging

Questions 38–47

38 Expand $5(5t - 8)$.

39 Use the distributive law to work out 3×999 .

40 $40 + 29 + 60 + 17$

from 5.01

41 Kai says $4(n + 3) = 4n + 3$. Is that right? Explain.

42 $\frac{2}{5}$ of 10

from 4.05

43 Work out $9 \times 26 + 9 \times 21$ the quick way.

44 Expand $5(5x - 9)$.

45 Write 11×8 another way. Is the answer the same?

from 5.01

46 Use the distributive law to work out 6×49 .

47 $5 \times 6 \times 2$

from 5.01

Challenging

Questions 48–57

48 Use the distributive law to work out 6×199 .

49 Ruby says $2(b + 8) = 2b + 8$. Is that right? Explain.

50 Write CCCXVI as a number.

from 3.06

51 Work out $4 \times 28 + 4 \times 26$ the quick way.

52 $87 + 24 + 3$

from 5.01

53 Expand $5(2p - 4)$.

54 Fill in the box:
 $4 \times 39 = 4 \times 30 + 4 \times \square$

55 Is 7×6 the same as 6×7 ?

from 5.01

56 Use the distributive law to work out 8×49 .

57 Write $2 \times 3 \times 3 \times 3$ in index form.

from 3.10

5.03 Variables and algebraic notation

YEAR 7 · CHAPTER 5A: ALGEBRA · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

A pronumeral (a letter) stands for a number. In algebra, leave out the $\times$ sign: $3 \times n = 3n$, $a \times b = ab$, $n \times n = n^2$. Write the number first, then the letters in alphabetical order.

Watch your teacher.
Copy every step.

E1 Simplify $5 \times y$.

E2 Simplify $b \times 3 \times a$.

E3 Matchstick squares in a row: 1 square uses 4, 2 squares use 7, 3 squares use 10. Write a rule.

Easy

Questions 1–11

1 Simplify $a \times 10$.

2 Simplify $t \times t$.

3 What does $4p$ mean?

4 Simplify $4 \times a$.

5 Simplify $c \times c$.

6 What does $12m$ mean?

7 Simplify $k \times 3$.

8 Simplify $a \times a$.

9 $47 + 57 + 3$

from 5.01

10 What does $8n$ mean?

11 Simplify $7 \times y$.

Easy

Questions 12–20

12 Simplify $p \times p$.

13 What does $11k$ mean?

14 Simplify $t \times 3$.

15 List the factors of 10.

from 3.09

16 Simplify $x \times b \times 7$.

17 Simplify $n \times n$.

18 What does $6a$ mean?

19 Work out $9 \times (10 + 2)$ two ways.

from 5.02

20 Simplify $b \times 3$.

Medium

Questions 21–28

21 Simplify $x \times c \times 5$.

22 Write $4ak$ in expanded form.

23 Is $(10 - 4) - 3$ the same as $10 - (4 - 3)$?

from 5.01

24 Write $t \div 6$ as a fraction.

25 The pattern is 4, 6, 8, 10, ...
Write a rule for term n .

26 Find the digit sum of 1 849. Is 1 849 divisible by 3?

from 3.04

27 Simplify $p \times k \times 3$.

28 Write $9kt$ in expanded form.

Medium

Questions 29–39

29 Write $5px$ in expanded form.

30 Write $m \div 5$ as a fraction.

31 Expand $4(p + 11)$.

from 5.02

32 The pattern is 8, 11, 14, 17, ...
Write a rule for term n .

33 $50 \times 13 \times 2$

from 5.01

34 Simplify $y \times n \times 4$.

35 Simplify $5 \times b \times n \times b \times 2$.

36 Write DXLV as a number.

from 3.06

37 Write $3mt$ in expanded form.

38 3×44

from 5.02

39 Write $m \div 6$ as a fraction.

Challenging

Questions 40–49

40 Simplify $3 \times n \times p \times n \times 5$.

41 A pattern has the rule
 $t = 2n + 4$. Find term 16.

42 $16 + 65 + 84 + 53$

from 5.01

43 Find the rule: when $x = 1, 2, 3, 4$,
 $y = 8, 12, 16, 20$.

44 Round 232 to its first digit.

from 3.01

45 Chloe says ty and yt are
different. Is that right?
Explain.

46 Simplify $3 \times a \times m \times a \times 2$.

47 Work out $8 \times (10 + 7)$
two ways.

from 5.02

48 A pattern has the rule
 $t = 5n + 1$. Find term 14.

49 Is $(8 + 5) + 2$ the same as
 $8 + (5 + 2)$?

from 5.01

Challenging

Questions 50–59

50 A pattern has the rule
 $t = 3n + 4$. Find term 9.

51 Find the rule: when $x = 1, 2, 3, 4$,
 $y = 7, 11, 15, 19$.

52 $525 \div 3$

from 3.03

53 Zac says kp and pk are
different. Is that right?
Explain.

54 Expand $2(a + 10)$.

from 5.02

55 Simplify $6 \times b \times c \times b \times 5$.

56 The pattern is 10, 14, 18, 22, ...
Write a rule for term n .

57 $5 \times 14 \times 2$

from 5.01

58 A pattern has the rule
 $t = 5n + 3$. Find term 20.

59 $\frac{3}{7}$ of 42

from 4.05

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Turn the words into maths. More than and sum: $+$. Less than and difference: $-$. Times and product: $\times$. Divided by: $\div$. Careful: "3 less than x " is $x - 3$.

Watch your teacher.
Copy every step.

E1 5 more than n

E2 3 less than twice x

E3 The cost of p pens at \$4 each

Easy

Questions 1–9

1 Write an expression: 4 more than a

2 Write an expression: 5 less than k

3 Write an expression: 6 times a

4 Write an expression: b divided by 8

5 Write an expression: the sum of a and 11

6 Write an expression: double k

7 Write an expression: b squared

8 Write an expression: c decreased by 10

9 Simplify $c \times c$.

from 5.03

Easy

Questions 10–18

10 Write an expression: 10 more than m

11 Write an expression: the product of b and m

12 Write an expression: t increased by 4

13 Is 2 a common factor of 14 and 30?

from 3.11

14 Write an expression: 3 less than twice b

first: $___b$

$= ___b - ___$

15 Write an expression: n divided by 8

16 Write an expression: 7 less than p

17 Work out $6 \times (30 + 4)$ two ways.

from 5.02

18 Write an expression: 9 times a

Medium

Questions 19–26

19 Write an expression: n squared, plus 2

20 Write an expression: 2 more than 5 times k

21 $11 + 31 + 89 + 84$

from 5.01

22 Write an expression: 6 times n , take away 4

23 Write an expression: a plus m , then divided by 7

24 $2 \times \frac{3}{5}$

from 4.06

25 Write an expression: 9 less than the product of k and m

26 Write an expression: the sum of c and 9, then times 9

Medium

Questions 27–37

- 27 Write an expression: 4 less than twice a

- 28 Write an expression: 4 more than 6 times b

- 29 What does $10a$ mean?

from 5.03

- 30 Write an expression: 8 times a , take away 9

- 31 Fill in the box:
 $5 \times 28 = 5 \times 20 + 5 \times \square$

from 5.02

- 32 Write an expression for the cost of n pens at \$14 each, plus \$17 for a folder.

- 33 Write an expression: m squared, plus 6

- 34 Write CCL as a number.

from 3.06

- 35 Write an expression: 8 less than the product of b and p

- 36 Write 5×11 another way. Is the answer the same?

from 5.01

- 37 Write an expression: the sum of a and 4, then times 9

Challenging

Questions 38–47

- 38 Write an expression for the cost of m pens at \$13 each, plus \$16 for a folder.

- 39 Write an expression for the number of legs on x spiders and 11 birds.

- 40 The pattern is 5, 8, 11, 14, ...
Write a rule for term n .

from 5.03

- 41 \$110 is shared equally by y people. Write an expression for each share.

- 42 $\frac{5}{6}$ of 30

from 4.05

- 43 Write in words: $2(m + 5)$

- 44 Write an expression for the cost of a pens at \$3 each, plus \$6 for a folder.

- 45 8×34

from 5.02

- 46 Write an expression for the number of legs on b spiders and 4 birds.

- 47 $125 \times 4 \times 8$

from 5.01

Challenging

Questions 48–57

- 48 Write an expression for the number of legs on t spiders and 8 birds.

- 49 \$110 is shared equally by p people. Write an expression for each share.

- 50 Find the LCM of 4, 6 and 5.

from 3.12

- 51 Write in words: $2(b + 5)$

- 52 Simplify $a \times a$.

from 5.03

- 53 Write an expression for the cost of m pens at \$2 each, plus \$5 for a folder.

- 54 Write an expression: 11 less than twice b

- 55 Expand $9(m - 8)$.

from 5.02

- 56 Write an expression for the number of legs on x spiders and 3 birds.

- 57 $61\,000 \div 1\,000$

from 3.03

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Substitute: replace each pronumeral with its number, then work it out with the order of operations. $3a + 2$ when $a = 5$ is $3 \times 5 + 2 = 17$.

Watch your teacher.
Copy every step.

E1 Find $4m - 3$ when $m = 6$.

E2 Find $ab + b$ when $a = 3$ and $b = 5$.

E3 $A = lw$. Find A when $l = 8$ and $w = 5$.

Easy

Questions 1–9

1 Find $n + 4$ when $n = 10$.

2 Find x^2 when $x = 8$.

3 Find $\frac{y}{4}$ when $y = 28$.

4 Find $2p$ when $p = 6$.

5 Find k^2 when $k = 4$.

6 Find $\frac{n}{3}$ when $n = 18$.

7 Find $n + 8$ when $n = 9$.

8 Find x^2 when $x = 9$.

9 Write an expression: 4 less than m

from 5.04

Easy

Questions 10–18

10 Find x^2 when $x = 4$.

11 Find $\frac{x}{8}$ when $x = 16$.

12 Find $5n$ when $n = 2$.

13 How many times does 12 go into 96?

from 3.05

14 Find ab when $a = 4$ and $b = 8$.

= ____ × ____

= ____

15 Find m^2 when $m = 3$.

16 Find $\frac{n}{3}$ when $n = 9$.

17 Simplify $n \times n$.

from 5.03

18 Find $8x + 1$ when $x = 9$.

Medium

Questions 19–26

19 Find $2a + b$ when $a = 2$ and $b = 9$.

20 $P = 2l + 2w$. Find P when $l = 10$ and $w = 5$.

21 $36 + 28 + 64 + 13$

from 5.01

22 Find $5n + 4$ when $n = -5$.

23 Find $3b - a$ when $a = 6$ and $b = 2$.

24 Expand $4(c - 8)$.

from 5.02

25 $P = 2l + 2w$. Find P when $l = 10$ and $w = 4$.

26 Find $2x + 3$ when $x = -6$.

Medium

Questions 27–37

- 27 $P = 2l + 2w$. Find P when $l = 5$ and $w = 12$.

- 28 Find $5x + 3$ when $x = -1$.

- 29 Evaluate 2^2 .

from 3.07

- 30 Find $a + 2b$ when $a = 5$ and $b = 3$.

- 31 Write 9×6 another way. Is the answer the same?

from 5.01

- 32 A plumber charges $C = 50h + 25$ dollars for h hours. Find the cost of a 5-hour job.

- 33 $P = 2l + 2w$. Find P when $l = 11$ and $w = 5$.

- 34 Write an expression: 8 times c

from 5.04

- 35 Find $4x + 4$ when $x = -6$.

- 36 Write $t \div 4$ as a fraction.

from 5.03

- 37 Find $5a - b$ when $a = 6$ and $b = 8$.

Challenging

Questions 38–47

- 38 A plumber charges $C = 56h + 20$ dollars for h hours. Find the cost of a 2-hour job.

- 39 $v = u + at$. Find v when $u = 10$, $a = 5$ and $t = 6$.

- 40 $1\,737 \div 9$

from 3.03

- 41 Find $a^2 - 3b$ when $a = -1$ and $b = 4$.

- 42 Expand $9(t + 8)$.

from 5.02

- 43 Ali says $3x^2 = 324$ when $x = 6$. Is that right? Explain.

- 44 A plumber charges $C = 57h + 17$ dollars for h hours. Find the cost of a 6-hour job.

- 45 $25 \times 19 \times 4$

from 5.01

- 46 $v = u + at$. Find v when $u = 9$, $a = 2$ and $t = 4$.

- 47 Write an expression: 4 less than twice c

from 5.04

Challenging

Questions 48–57

- 48 $v = u + at$. Find v when $u = 5$, $a = 2$ and $t = 10$.

- 49 Find $a^2 - 3b$ when $a = -4$ and $b = 6$.

- 50 List the factors of 18.

from 3.09

- 51 Lena says $3x^2 = 225$ when $x = 5$. Is that right? Explain.

- 52 $58 + 45 + 2$

from 5.01

- 53 A plumber charges $C = 38h + 33$ dollars for h hours. Find the cost of an 8-hour job.

- 54 Find $5n + 6$ when $n = -1$.

- 55 Simplify $t \times 8$.

from 5.03

- 56 $v = u + at$. Find v when $u = 11$, $a = 2$ and $t = 4$.

- 57 29×41

from 5.02

YEAR 7 · CHAPTER 5A: ALGEBRA · PAGE 1 OF 2

NAME _____

CLASS

DATE _____

KEY IDEA

Watch your teacher.
Copy every step.

E1 $5x + 3x$

E2 $7a + 2b - 3a$

E3 Are $3xy$ and $5yx$ like terms?

Questions 1-9

1 Simplify $11t + t$.

2 Simplify $14a - a$.

3 Are $5k^2$ and $5k$ like terms?

4 Simplify $5n + 11n$.

5 Simplify $6x - 4x$.

6 Are $3a$ and $3p$ like terms?

7 Simplify $9p + 2p$.

8 Simplify $9p - 2p$.

9 Find x^2 when $x = 4$.

from 5.05

Questions 10–18

10 Simplify $9x - 3x$.

11 Are $3c^2$ and $3c$ like terms?

12 Simplify $12n + 6n$.

13 How many times does 25 go into 175?

from 3.05

14 Simplify $6c + 7m + 7c + m$.

$$= \underline{\hspace{1cm}} c + \underline{\hspace{1cm}} c + \underline{\hspace{1cm}} m + m$$

$$= \text{---} C + \text{---} m$$

15 Simplify $13t - 8t$.

16 Are $7t^2$ and $7t$ like terms?

17 Simplify $a \times a$.

from 5.03

18 Simplify $7k + 8k$.

Questions 19–26

19 Simplify $4m + 8y + 5m - 6y$.

20 A rectangle is $4p$ long and $9y$ wide. Write its perimeter as simply as possible.

21 $19 + 13 + 81 + 27$

from 5.01

22 Simplify $4c + 4 + 7c$.

23 Simplify $8k + 7x + 7k - x$.

24 Write an expression: 5 more than 8 times t

from 5.04

25 A rectangle is $8a$ long and $5x$ wide. Write its perimeter as simply as possible.

26 Simplify $3a + 8 + 6a$.

Medium

Questions 27–37

- 27 A rectangle is $7b$ long and $6p$ wide. Write its perimeter as simply as possible.

- 28 Simplify $3b + 3 + 3b$.

- 29 $\sqrt[3]{8}$

from 3.02

- 30 Simplify $7b + 3y + b + y$.

- 31 Work out $9 \times (10 + 2)$ two ways.

from 5.02

- 32 Simplify $6am + 6a^2 - am + 3a^2 + 3$.

- 33 A rectangle is $4t$ long and $3y$ wide. Write its perimeter as simply as possible.

- 34 Find $\frac{x}{4}$ when $x = 16$.

from 5.05

- 35 Simplify $8p + 6 + 7p$.

- 36 Write $y \div 2$ as a fraction.

from 5.03

- 37 Simplify $2a + 4p + 6a + 5p$.

Challenging

Questions 38–47

- 38 Simplify $7cx + 3c^2 - 6cx + 6c^2 + 9$.

- 39 Simplify $2b - 6n - 13b + 3n$.

- 40 $3\frac{1}{2} \times \frac{1}{2}$

from 4.06

- 41 Mia says $5c + 4m = 9cm$. Is that right? Explain.

- 42 Write an expression: 5 times b

from 5.04

- 43 A triangle has sides $3b$, $5b$ and $6b$. Write its perimeter, then find it when $b = 4$.

- 44 Simplify $6bk + 8b^2 - 4bk + b^2 + 6$.

- 45 Is $12 \div 4$ the same as $4 \div 12$?

from 5.01

- 46 Simplify $9c - 3y - 15c + 4y$.

- 47 Find ab when $a = 5$ and $b = 9$.

from 5.05

Challenging

Questions 48–57

- 48 Simplify $7a - 7x - 10a + 4x$.

- 49 Ali says $4b + 4c = 8bc$. Is that right? Explain.

- 50 Which is larger: $\frac{1}{10}$ or $\frac{1}{7}$?

from 4.02

- 51 A triangle has sides $3c$, $4c$ and $4c$. Write its perimeter, then find it when $c = 4$.

- 52 Expand $7(x + 7)$.

from 5.02

- 53 Simplify $9cp + 6c^2 - cp + 7c^2 + 3$.

- 54 Simplify $9y + 9 + y$.

- 55 Simplify $t \times 6$.

from 5.03

- 56 Simplify $7m - 7y - 10m + 8y$.

- 57 Write an expression: 12 less than twice p

from 5.04

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To multiply terms, multiply the numbers, then the letters. Write the letters in alphabetical order and use index notation: $3a \times 4a = 12a^2$.

Watch your teacher.
Copy every step.

E1 $4 \times 3y$

E2 $5a \times 2b$

E3 $3m \times (-2m)$

Easy

Questions 1–16

1 $8 \times 2n$

2 $7m \times 4x$

3 $7n \times n$

4 $2 \times 9a$

5 $5m \times 4n$

6 $5p \times p$

7 $7 \times 6n$

8 $4n \times 3t$

9 Simplify $15m - 11m$.

from 5.07

10 $2y \times y$

11 $8 \times 6c$

12 $5n \times 4t$

13 $6x \times x$

14 $5 \times 2k$

15 $6c \times 2m$

16 $4y \times y$

Easy

Questions 17–27

17 $3t \times 4y$

18 $9t \times t$

19 $4 \times 8m$

20 Evaluate 7^2 .

from 5.07

21 $6p \times 3t$

$= ___ \times ___ \times p \times t$

$= ___ pt$

22 $5m \times 6y$

23 $3b \times b$

24 Simplify $t \times t$.

from 5.03

25 $5 \times 2c$

26 $3c \times 4n$

27 $5b \times 3n$

Medium

Questions 28–35

28 $6ky \times 2ky$

29 A rectangle is $2t$ cm long and $9y$ cm wide. Write its area.

30 $14 + 58 + 86 + 37$

from 5.01

31 Simplify $2x \times 2x$.

32 $4ac \times 2a$

33 $P = 2l + 2w$. Find P when $l = 11$ and $w = 5$.

from 5.05

34 A rectangle is $3n$ cm long and $7t$ cm wide. Write its area.

35 Simplify $2y \times 2y$.

Medium

Questions 36–46

- 36 A rectangle is $7b$ cm long and $2p$ cm wide. Write its area.

- 37 Simplify $4t \times 4t$.

- 38 What is 10% of 260?

from 4.11

- 39 $5c \times 3ck$

- 40 Work out $7 \times (40 + 3)$ two ways.

from 5.02

- 41 $-3t \times 5y \times 3t$

- 42 A rectangle is $3m$ cm long and $4x$ cm wide. Write its area.

- 43 Write an expression: 5 less than k

from 5.04

- 44 Simplify $4a \times 4a$.

- 45 Simplify $5y + 7 + y$.

from 5.07

- 46 $5np \times 6n$

Challenging

Questions 47–56

- 47 $-3b \times (-6p) \times 3b$

- 48 Simplify $3a \times 3a \times 3a$.

- 49 $5\frac{1}{4} + 2\frac{2}{3}$

from 4.04

- 50 Mia says $3a \times 4a = 12a$. Is that right? Explain.

- 51 What does $5x$ mean?

from 5.03

- 52 $-6b \times 2m \times 2b$

- 53 Simplify $2b \times 2b \times 2b$.

- 54 Is $(10 - 4) - 3$ the same as $10 - (4 - 3)$?

from 5.01

- 55 Tom says $5k \times 6k = 30k$. Is that right? Explain.

- 56 Find $3n + 3$ when $n = -2$.

from 5.05

Challenging

Questions 57–66

- 57 Simplify $2a \times 2a \times 2a$.

- 58 $6m \times (-6y) \times 2m$

- 59 $\frac{7}{9} - \frac{3}{9}$

from 4.03

- 60 Simplify $3a \times 3a$.

- 61 Expand $7(c + 3)$.

from 5.02

- 62 Simplify $4b \times 4b \times 4b$.

- 63 $-3m \times (-4x) \times 3m$

- 64 Write an expression: 4 times k

from 5.04

- 65 Simplify $6m \times 6m$.

- 66 Simplify $6k + 2m + 3k - 7m$.

from 5.07

5.09 Dividing terms

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

To divide terms, write the division as a fraction, then cancel the numbers and the letters: $12ab \div 4a = 3b$.

Watch your teacher.
Copy every step.

E1 $15x \div 3$

E2 $12ab \div 4a$

E3 $\frac{18m^2}{6m}$

Easy

Questions 1–16

1 $21p \div 3$

2 $kx \div k$

3 $5t \div t$

4 $20y \div 5$

5 $mn \div m$

6 $12p \div p$

7 $40m \div 8$

8 $bc \div b$

9 $3b \times 4c$

from 5.08

10 $2a \div a$

11 $48b \div 8$

12 $bt \div b$

13 $11a \div a$

14 $72t \div 9$

15 $ay \div a$

16 $10t \div t$

Easy

Questions 17–27

17 $px \div p$

18 $3b \div b$

19 $8m \div 2$

20 Find the digit sum of 3 714. Is 3 714 divisible by 3?

from 3.04

21 $32mp \div 4p$

$= \frac{mp}{p}$

$= m$

22 $am \div a$

23 $7m \div m$

24 Simplify $13a - 9a$.

from 5.07

25 $24n \div 8$

26 $12ab \div 3b$

27 $mp \div m$

Medium

Questions 28–35

28 $24mn \div 4m$

29 Simplify $\frac{9p}{3}$.

30 $23 + 14 + 77 + 40$

from 5.01

31 Simplify $\frac{6kp}{3p}$.

32 $10by \div 2y$

33 $P = 2l + 2w$. Find P when $l = 13$ and $w = 9$.

from 5.05

34 Simplify $\frac{36b}{4}$.

35 Simplify $\frac{10mt}{2t}$.

Medium

Questions 36–46

36 Simplify $\frac{10t}{5}$.

37 Simplify $\frac{9mn}{3n}$.

38 Which is larger: $\frac{1}{4}$ or $\frac{1}{6}$?

from 4.02

39 $8bn \div 2n$

40 Work out $7 \times (10 + 4)$ two ways.

from 5.02

41 Simplify $\frac{6c2y}{2cy}$.

42 Simplify $\frac{6t}{3}$.

43 $3c \times c$

from 5.08

44 Simplify $\frac{15kn}{5n}$.

45 Simplify $3c + 4 + 5c$.

from 5.07

46 $18ax \div 2x$

Challenging

Questions 47–56

47 Simplify $\frac{20p2x}{4px}$.

48 A rectangle has area $32py$ and length $4p$. Find its width.

49 $5\frac{1}{2} + 2\frac{2}{5}$

from 4.04

50 $-9n \div 3n$

51 Find $\frac{n}{6}$ when $n = 24$.

from 5.05

52 Simplify $\frac{18k2t}{6kt}$.

53 A rectangle has area $63at$ and length $7a$. Find its width.

54 Write $2kp$ in expanded form.

from 5.03

55 $-15m \div 3m$

56 $3ck \times 2k$

from 5.08

Challenging

Questions 57–66

57 A rectangle has area $27xy$ and length $9x$. Find its width.

58 $-16x \div 4x$

59 Write 6 out of 9 as a fraction in simplest form.

from 4.12

60 Simplify $\frac{30c2n}{6cn}$.

61 Write an expression: 2 less than b

from 5.04

62 Simplify $\frac{8cx}{4x}$.

63 A rectangle has area $48at$ and length $6a$. Find its width.

64 Simplify $9k + 11k$.

from 5.07

65 $-10t \div 2t$

66 Find ab when $a = 5$ and $b = 4$.

from 5.05